

A Chinese power text classification algorithm based on deep active learning

Song Deng^{a,*}, Qianliang Li^a, Renjie Dai^b, Siming Wei^b, Di Wu^c, Yi He^d, Xindong Wu^e

^a Institute of Advanced Technology, Nanjing University Post & Telecommunication, 210003, Nanjing, China

^b State Grid Shanghai Municipal Electric Power Company, 200122, Shanghai, China

^c College of Computer and Information Science, Southwest University, Chongqing, 400715, China

^d Old Dominion University, Norfolk, 23462, VA, United States of America

^e Key Laboratory of Knowledge Engineering with Big Data (the Ministry of Education of China), Hefei University of Technology, Hefei, 230009, China

ARTICLE INFO

Keywords:

Natural language processing
Deep active learning
Hierarchical confidence
Power text
Knowledge graph

ABSTRACT

The construction of knowledge graph is beneficial for grid production, electrical safety protection, fault diagnosis and traceability in an observable and controllable way. Highly-precision text classification algorithm is crucial to build a professional knowledge graph in power system. Unfortunately, there are a large number of poorly described and specialized texts in the power business system, and the amount of data containing valid labels in these texts is low. This will bring great challenges to improve the precision of text classification models. To offset the gap, we propose a classification algorithm for Chinese text in the power system based on deep active learning (CCTP-DAL). Our core idea is to apply a hierarchical confidence strategy to a deep active learning model, to balance the trade-offs between the amount of training data and the accuracy of text classification. Our CCTP-DAL (1) trains the Bert model using a small amount of labeled data to calculate the confidence level of each short text, (2) selects high-confidence text data with optimal model generalization capability based on the hierarchical confidence level, and (3) fuses deep learning models and active learning strategies to ensure high text classification accuracy with less labeled training data. We benchmark our model on a real crawler data on the web with extensive experiments. The experimental results demonstrate that our proposed model can achieve higher text classification accuracy with less labeled training data compared with other deep learning models.

1. Introduction

Advanced information and communication technologies have driven the power system into the digital power era [1], which makes the knowledge system in the power field increasingly present the trend of openness, flatness and boundary blurring. Knowledge in power system presents complex and variable characteristics. At the same time, the heterogeneous knowledge is increasing and the update speed of knowledge is accelerated [2]. However, the existing knowledge organization and management methods cannot meet the application of the growing power system knowledge. Knowledge graph is a kind of graph database that integrates artificial intelligence and traditional database. Building a knowledge graph in the electric power domain enable the electric power system the ability to mine and analyze useful knowledge from large-scale textual information, which can connect the scattered knowledge points in the electric power domain. Furthermore, it can also enhance the information retrieval ability and assist in power grid decision-making [3,4].

The rapid progress and development of power systems generate huge amounts of power text data every day. These text data are the basis for building a knowledge graph in the electric power domain. Unlike general domain texts, power domain texts contain many mathematical symbols, physical mechanisms, and summaries of engineering experience. With few descriptive vocabularies and weak descriptive information, they can be characterized as high-dimensional, sparse and unclear topic expressions [5,6]. In addition to technical reports containing mathematical symbols, physical mechanisms, and summaries of engineering experience, as well as structured data, there is also a lot of text data in the power system. These text data mainly come from power operation rules [7], operation and maintenance records [8], power market [9], fault diagnose [10], auditing [11] and other fields. How to mine business knowledge related to power operation and maintenance, fault diagnosis, market analysis, auditing, etc. from these textual data through text classification is a crucial part in constructing a knowledge graph in the power domain. High-precision power text classification

* Corresponding author.

E-mail addresses: dengsong@njupt.edu.cn (S. Deng), rxzzzleo@gmail.com (Q. Li), dairj@sh.sgcc.com.cn (R. Dai), weisiming1990@126.com (S. Wei), wudi1986@swu.edu.cn (D. Wu), yihe@cs.odu.edu (Y. He), xwu@hfut.edu.cn (X. Wu).

<https://doi.org/10.1016/j.asoc.2023.111067>

Received 27 March 2023; Received in revised form 9 November 2023; Accepted 16 November 2023

Available online 22 November 2023

1568-4946/© 2023 Elsevier B.V. All rights reserved.

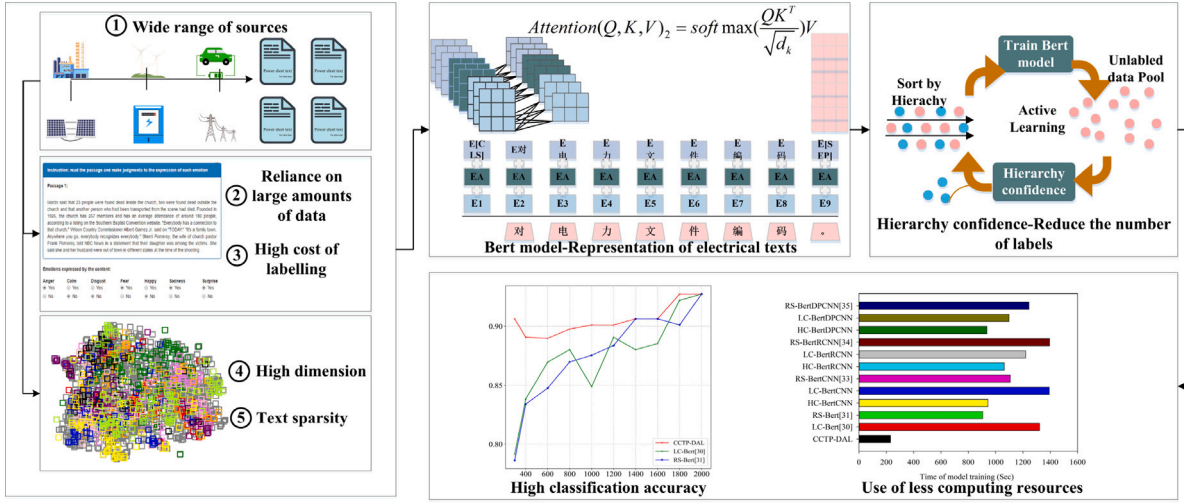


Fig. 1. The framework of our research.

algorithms are essential for building power knowledge graphs. Current text classification methods include traditional machine learning-based text classification methods and deep learning-based text classification methods. Traditional machine learning-based text classification methods rely on the manual formulation of text feature extraction rules [12] and they then generate domain-specific text dictionaries. However, traditional machine-learning-based text classification methods are weak in expressing text features and do not represent the logical relationships between texts well. Deep learning-based text classification methods do not require the manual extraction of text features. A deep learning model is used to encode the text vector, and the accuracy of text classification is improved by increasing the size of the training data and the number of iterations. However, deep learning-based text classification methods are highly reliant on the number of labeled texts. For the power sector, text labeling is too costly to train efficient and highly accurate text classification models [13].

To ensure highly precise text classification under the conditions of a few of text annotations in the electricity domain, this paper combines active learning mechanisms with deep learning methods and proposes a deep active learning-based text classification method for electricity. The method can exploit the advantages of deep learning in both high-dimensional data processing and automatic feature extraction. It also combines the great potential of the active learning mechanism in effectively reducing the cost of data annotation. This ultimately allows our proposed model to obtain high text classification accuracy with a low number of text annotations. The framework of our research is shown in Fig. 1.

The main contributions of this paper are listed as follows.

- To address the problem of insufficient labeled data in the electric power domain, we propose an active learning algorithm based on hierarchical confidence, which is effective in reducing the dependence of text classification algorithms on large-scale labeled data.
- We propose a deep active learning-based text classification algorithm for power system to ensure text classification accuracy under the condition that the labeled data quantity is limited.
- Experimental results on real network data indicate that our proposed algorithm has higher text classification accuracy and efficiency under the condition of small amounts of annotations as compared with the state-of-the-art text classification algorithms based on deep learning.

The remainder of this paper is organized as follows. Section 2 introduces the related work on text classification for electricity. Section 3 analyzes the short text classification algorithm based on deep

active learning. Section 4 conducts the comparative experiments on real network data and analyzes the results. The conclusions and highlights are summarized in Section 5.

2. Related work

This section provides an overview of existing models and existing algorithms for text classification. The advantages and disadvantages of these modeling approaches and their relevance to our recommendations are discussed. In general, current text classification algorithms can be divided into traditional machine learning-based methods and deep learning-based methods, as explained in the following discussion.

2.1. Traditional machine learning-based text classification algorithms

The common methods used in traditional text classification methods are NB algorithm, SVM algorithm, KNN algorithm, DT decision tree, etc. The plain Bayesian algorithm achieves the classification of text based on posterior probability [14]. Klopotek et al. [15] proposed an improved Bayesian text classification algorithm to improve the classification effect of the Bayesian model as the text vocabulary increases. The traditional machine learning text classification method is poor in characterizing high-dimensional text feature data. Therefore, Kim et al. [16] proposed an improved SVM text classification algorithm to reduce the input dimension of the model based on the center-of-mass dimension reduction method. Expect the commonly used algorithms above, decision tree models are also widely used in text classification due to their advantage of good dimension reduction of text features. Su et al. [17] proposed an improved C4.5 decision tree model text classification algorithm. Unsupervised learning does not rely on labeled training samples, Yong et al. [18] proposed an improved K-means text classification algorithm. The performance improvement of a single classification model is limited and is inspired by the idea of integration learning. Shi et al. [19] proposed a chapter on semi-supervised learning algorithms based on tolerance rough sets and integrated learning. Using tolerance rough set theory to obtain initial counterexample samples of sample data, the NB classifier converges to the positive and negative class true boundaries of the feature space. Although the traditional machine learning-based text classification algorithms have achieved good text classification results, the machine learning-based text classification methods need to design different text features and select different classifiers according to different tasks, which is highly subjective when specifying feature rules manually. And most of the text classification methods based on machine learning methods are based on bag-of-words models, which limits the performance of classifiers [20].

2.2. Deep learning-based text classification algorithms

Compared to traditional machine learning methods, deep learning text classification methods can improve the representational power of text without relying on manually developed feature extraction rules. Also, trained text features can be applied to new text classification tasks by using transfer learning methods. The model's ability to extract corpora of complex text was improved by combining attention mechanisms [21–23]. The problem of power fault text classification was solved based on LSTM text classification model to help regulators better understand and control the risks in power system operation [24,25]. To improve the generalization ability of natural language models, Google Research proposed the Bert pre-training model [26], based on Transformer feature extraction for deep characterization of contextual contexts. The use of fine-tuning strategies to solve different downstream tasks provides a better approach for solving natural language processing tasks. The Bert model was expanded by combining the GAN network [27,28]. Chai et al. [29] proposed a new strategy to combine Bert model with reinforcement learning to improve the classification effect. To reduce the computational complexity of text classification algorithms based on the Bert model, Wen et al. [30] proposed an optimization method based on matrix decomposition and parameter sharing to reduce the computational complexity of the Bert model. Although deep learning achieves better results in text classification, the accuracy of the model relies on the size of the training dataset, which limits the application of text classification algorithms in practical scenarios.

2.3. Remark and our thoughts

The general process of text classification methods is to pre-process the text, followed by vectoring the processed text. Feature extraction is performed on the text vector, and finally, automatic classification of text is achieved based on the text features. However, traditional machine learning-based text classification algorithms are weak in characterizing text and require the construction of a dictionary to solve specialized text classification tasks. They can achieve good classification results for specific tasks, but the generalization of the model is too poor and the model is not portable. In addition, deep learning-based text classification algorithms are better at characterizing text and using deep learning models to automatically extract text features, but the model parameters are complicated and overly dependent on large-scale labeled datasets. Different from text classification in general-purpose fields, text for the power sector is highly specialized and lacks large-scale annotated data. Therefore, this paper proposes (1) an active learning strategy based on hierarchical confidence, and (2) combines the hierarchical confidence strategy with a deep learning-based text classification algorithm to reduce the dependence of the text classification algorithm on annotated data and improve the accuracy of power text classification under the condition of small amount of data labeled.

3. Power text classification algorithm based on deep active learning

In this section, the natural language model BERT, and the hierarchical confidence level active learning strategy, will be introduced in detail. The active learning strategy combined with a natural language model is examined to improve the classification ability of the model when only a small amount of training data is available.

3.1. Deep active learning-based power classification model

With the widespread use of new information and communication technologies such as edge computing and 5G in new power systems, text data around the fields of power generation, transmission, transformation, distribution, consumption, and dispatch are becoming increasingly abundant [31–33]. Highly accurate text classification is crucial for building grid-oriented expertise graphs. In recent years, text classification algorithms based on deep learning have received much attention [18,23,24]. However, deep learning-based text classification algorithms are too dependent on the number of labeled texts. The cost of text labeling is too high to train efficient and highly accurate text classification models. In order to reduce the dependence of deep learning-based text classification methods on large-scale training data, this paper combines deep learning models with hierarchical confidence active learning mechanisms. A deep active learning-based power text classification algorithm (CCTP-DAL) is proposed. A schematic diagram of the algorithm is shown in Fig. 2.

In the deep active learning-based power text classification algorithm (CCTP-DAL), we calculate the confidence level of each power text vocabulary by using the Gelu activation function. A small amount of unlabeled data is selected for labeling according to the active learning strategy of hierarchical confidence. Training the BERT power text classification model based on a small amount of labeled text data reduces the dependence of the deep learning model on large-scale data. Finally, the trained power text classification model is used to achieve automatic classification of power texts based on the migration learning idea.

3.2. Bert

BERT is a deep bidirectional language representation model based on a transformer. A multilayer bidirectional encoder network is constructed by using transformer structure. We can divide the BERT model into three parts, the embedding of the model, the transformer block, and the classification layer. The model structure is shown in Fig. 3.

(1) Embedding

The encoding of electrical texts is the foundation of solving the problem of classifying electrical texts. $T\{T_1, T_2, T_3, \dots, T_N\}$ represents power text $E\{E_{T1}, E_{T2}, E_{T3}, \dots, E_{TN}, E_{S1}, E_{S2}, E_{S3}, \dots, E_{SN}, E_{P1}, E_{P2}, E_{P3}, \dots, E_{PN}\}$ represents the encoding of the power text. The encoding of power text consists of token embedding, segment Embedding and position Embedding. The vector representation of the final input model is summed by their corresponding positions. The first position (CLS) encodes the information of the whole sentence that can be used for classification and (SEP) separates two sentences. Segment embedding encodes 0 and 1 to between two different sentences in a paragraph. Position embedding encodes the position of the corresponding word.

(2) Transformer block

Two of the most distinctive features of the transformer block are self-attention and Multi-head Self-Attention. In the same way that humans focus their attention on target areas when acquiring information, attention mechanisms can quickly sift through vast amounts of complex information to find the valuable information needed. The attention mechanism is mainly represented by the query vector (Q), the key vector (K) and the value vector (V). Each of the three vectors is calculated by the following equations.

$$Q = W^Q E \quad (1)$$

$$K = W^K E \quad (2)$$

$$V = W^V E \quad (3)$$

where W^Q is the generating matrix corresponding to Q , W^K is the generating matrix corresponding to K , W^V is the generating matrix corresponding to V , and E represents the encoding of the power text.

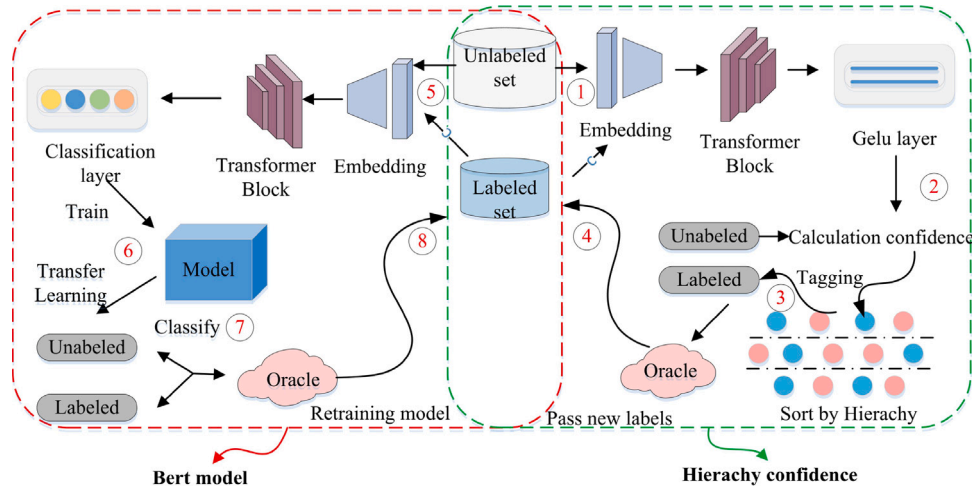


Fig. 2. Schematic diagram of power text classification algorithm based on deep active learning.

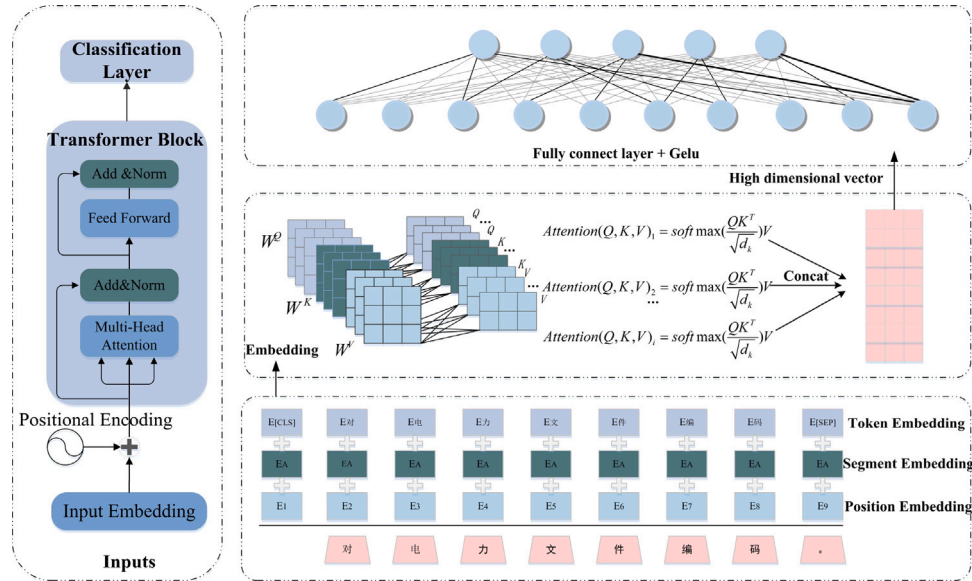


Fig. 3. BERT model structure.

The attention value of the power text can be calculated by using the three vectors of Q , K and V . The attention mechanism Eq. (4) is calculated as follows.

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

where d_k is the dimensionality of the K vector.

To further improve the extraction the power text information, multihead self-attention (MSA) is an attention mechanism that is repeated h times in separate heads. The linearly varying parameter matrices W are different for different heads, and then the results of h times are stitched together and linearly transformed as the output of the multihead attention mechanism.

$$head_i = Attention(Q_i, K_i, V_i) \quad (5)$$

$$Multihead(Q, K, V) = \text{Concat}(head_1, head_2, \dots, head_i)W^0 \quad (6)$$

where $head_i$ denotes the result of the attention mechanism calculation under different headers. Q_i, K_i, V_i denotes the three vectors of Q, K and V calculated under different headers corresponding to the generation matrix, and W^0 is the output weight matrix.

The purpose of this mechanism is to learn the dependencies between elements within a sequence and to mine the internal structural features of the power text. Compared with traditional CNN or RNN models, it solves the problem of long-distance dependencies.

(3) Classification layer

The classification of a power text is a typical nonlinear classification process, which is achieved by using a fully connected network and the Gelu activation function to reduce the dimensionality and automatically classify high-dimensional text feature vectors. Hendrycks et al. [34] demonstrated that Gelu is a more suitable activation function for natural language processing tasks. The Gelu activation function is shown in Eq. (7).

$$GELU(x) = xP(X \leq x) = x\Phi(x) \quad (7)$$

where $\Phi(x)$ conforms to the normal distribution probability function $X : N(0, 1)$.

3.3. Hierarchy confidence

Active learning strategies can be used to improve the accuracy of text classification algorithms by obtaining larger values of labeled

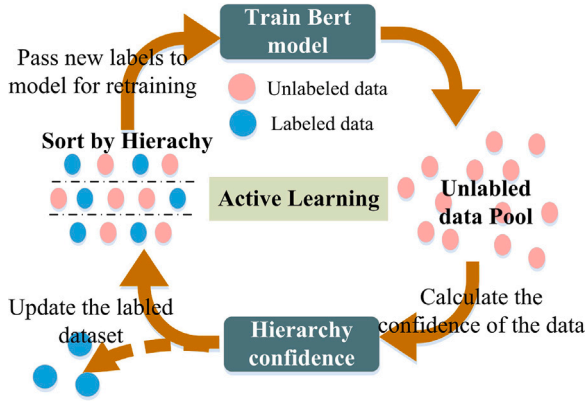


Fig. 4. Active learning strategy with hierarchical confidence.

data at a lower cost. The traditional least Confidence based active learning strategy (Least Confidence) selects data that are most difficult for the model to identify and classify for annotation. The least confidence active learning strategy focuses too much on low-confidence sample data, and the variability between the selected data samples is insufficient to reflect the diversity of the data features, which is not conducive to the performance improvement of the model. To adapt to the development of deep learning models and improve the model's ability to perceive data diversity, we propose an active learning strategy based on hierarchy confidence (Hierarchy Confidence).

Definition 1. Let $X = \{X_1, X_2, \dots, X_N\}$ denote the power text dataset, $X_i, i = \{1, 2, \dots, N\}$ denote the i th power text, $L_j, j = \{1, 2, \dots, K\}$ denote the category of the power text predicted by the classification model, and $Y_{ij} = \text{Gelu}(L_j | X_i)$ denote the probability that the i th power text belongs to the j th category. Then $C_i = \max(Y_{ij})$ is called the confidence level of the i th power text.

Definition 2. Let $X_i^h, i = \{1, 2, \dots, N\}, h = \{1, 2, \dots, H\}$ denote the i th power text in the h th layer, H refer the number of layers into which the sample dataset X is divided, N_h be the number of all power texts in the h th layer, K_h be the sampling probability of each layer, $L_j, j = \{1, 2, \dots, k\}$ denote the category of the power text predicted by the classification model, and $Y_{ij} = \text{Gelu}(L_j | X_i^h)$ denote the probability that the i th power text in the h th layer belongs to the j th category. Then, we call that $C_i^h = \max(Y_{ij})$ is the confidence level of the i th power text of the h th layer, and x_{HC}^* is the hierarchical confidence.

Where x_{HC}^* is shown in Eq. (8).

$$x_{HC}^* = \sum_{h=1}^H \sum_{i=1}^{K_h * N_h} \sum_{j=1}^k \argmin(\max(\text{Gelu}(L_j | X_i^h)))$$

$$= \sum_{h=1}^H \sum_{i=1}^{K_h * N_h} \argmin(C_i^h) \quad (8)$$

The hierarchical confidence active learning strategy ranks the text data according to their confidence level. The power text data are divided into several layers. Text with small confidence differences are grouped into one layer, and power text data with large confidence differences are grouped into different layers. The aim is to have as large as possible confidence differences in the power text in different layers and less variability in the power text within the same layer. The active learning strategy for hierarchical confidence is shown in Fig. 4.

Definition 3. Let $X = \{X_1, X_2, \dots, X_N\}$ be a text data set, $X_i, i = \{1, 2, \dots, N\}$ denote the i th power text, $C_i, i \in [1, N]$ denote the confidence level of the i th text data in X , and $\bar{C} = \sum_{i=1}^N C_i$ denotes the average confidence levels of all data in X . Then, $S^2 = \frac{1}{N-1} \sum_{i=1}^N (C_i - \bar{C})^2$ is called the confidence variance of X .

Theorem 1. The data set $X' = \{X'_1, X'_2, \dots, X'_n\}$ is formed by randomly selecting n text data from X , and $f = \frac{n}{N}$ is the proportion of randomly selected power texts from X . Then, the mean variance of the dataset X' is $V(\bar{C}) = \frac{1-f}{n} S^2$.

Proof. We first introduce the random variable $a_i, i \in [1, n]$, if X_i is selected, then $a_i = 1$; and vice versa, $a_i = 0$. From Definition 3, we know that $\bar{C} = \sum_{i=1}^N C_i$, then the following equation holds

$$\bar{C} = \frac{1}{n} \sum_{i=1}^N a_i C_i. \quad (9)$$

From Eq. (9), we have

$$V(\bar{C}) = V\left(\frac{1}{n} \sum_{i=1}^N a_i C_i\right). \quad (10)$$

And, we let $\text{Cov}(X, Y)$ denote the covariance between the two random variables X and Y . Then the Eq. (10) can be written as

$$V(\bar{C}) = \frac{1}{n^2} \left(\sum_{i=1}^N C_i^2 V(a_i) + 2 \sum_{i < j} C_i C_j \text{cov}(a_i, a_j) \right). \quad (11)$$

According to the covariance formula, we have

$$\begin{aligned} V(\bar{C}) &= \frac{1}{n^2} \left(\sum_{i=1}^N C_i^2 V(a_i) + 2 \sum_{i < j} C_i C_j \text{cov}(a_i, a_j) \right) \\ &= \frac{1}{n^2} \left\{ \sum_{i=1}^N C_i^2 \frac{n}{N} \frac{N-n}{N} + 2 \sum_{i < j} C_i C_j \left[\frac{n}{N(N-1)} \left(1 - \frac{n}{N} \right) \right] \right\} \\ &= \frac{1}{n^2} \left\{ \sum_{i=1}^N C_i^2 \frac{n}{N} \frac{N-n}{N} + 2 \sum_{i < j} C_i C_j \left[-\frac{f(1-f)}{N-1} \right] \right\} \\ &= \frac{1}{n^2} \left\{ \sum_{i=1}^N C_i^2 \frac{n}{N} (1-f) + 2 \sum_{i < j} C_i C_j \left[-\frac{n}{N} \frac{1-f}{N-1} \right] \right\} \\ &= \frac{1}{n^2} \frac{n}{N} (1-f) \left[\sum_{i=1}^N C_i^2 - 2 \frac{1}{N-1} \left(\sum_{i < j} C_i C_j \right) \right] \\ &= \frac{1-f}{nN} \left[\frac{N}{N-1} \sum_{i=1}^N C_i^2 - \frac{1}{N-1} \sum_{i=1}^N Y_i^2 - 2 \frac{1}{N-1} \left(\sum_{i < j} C_i C_j \right) \right] \\ &= \frac{1-f}{nN} \left[\frac{N}{N-1} \sum_{i=1}^N C_i^2 - \frac{1}{N-1} \left(\sum_{i=1}^N Y_i^2 + 2 \left(\sum_{i < j} C_i C_j \right) \right) \right] \\ &= \frac{1-f}{nN} \left[\frac{N}{N-1} \sum_{i=1}^N C_i^2 - \frac{1}{N-1} \left(\sum_{i=1}^N Y_i \right)^2 \right] \\ &= \frac{1-f}{n(N-1)} \left[\sum_{i=1}^N C_i^2 - N \left(\frac{1}{N} \sum_{i=1}^N Y_i \right)^2 \right] \\ &= \frac{1-f}{n(N-1)} \sum_{i=1}^N (C_i^2 - \bar{C}^2) \end{aligned} \quad (12)$$

Finally, from Definition 3, we have

$$V(\bar{C}) = \frac{1-f}{n(N-1)} \sum_{i=1}^N (C_i^2 - \bar{C}^2) = \frac{1-f}{n} S^2. \quad (13)$$

Definition 4. Let the power text data set X be divided into L layers, N_h be the number of all power texts in the h th layer, $X_i^h, i = \{1, 2, \dots, N_h\}, h = \{1, 2, \dots, L\}$ denote the i th power text in the h th layer, and $\sum_{h=1}^L N_h = N$. $C_i^h = \max(y_{ij})$ means the confidence level of the i th power text in the h th layer, and $\bar{C}_h = \sum_{i=1}^{N_h} C_i^h$ denote the average confidence level of all power texts in the h th layer. Then, $S_h^2 = \frac{1}{N_h-1} \sum_{i=1}^{N_h} (C_i^h - \bar{C}_h)^2$ is said to be the hierarchical confidence variance of X .

Theorem 2. Based on the hierarchical sampling, n data are selected from the power text dataset X to form the power text sampling dataset

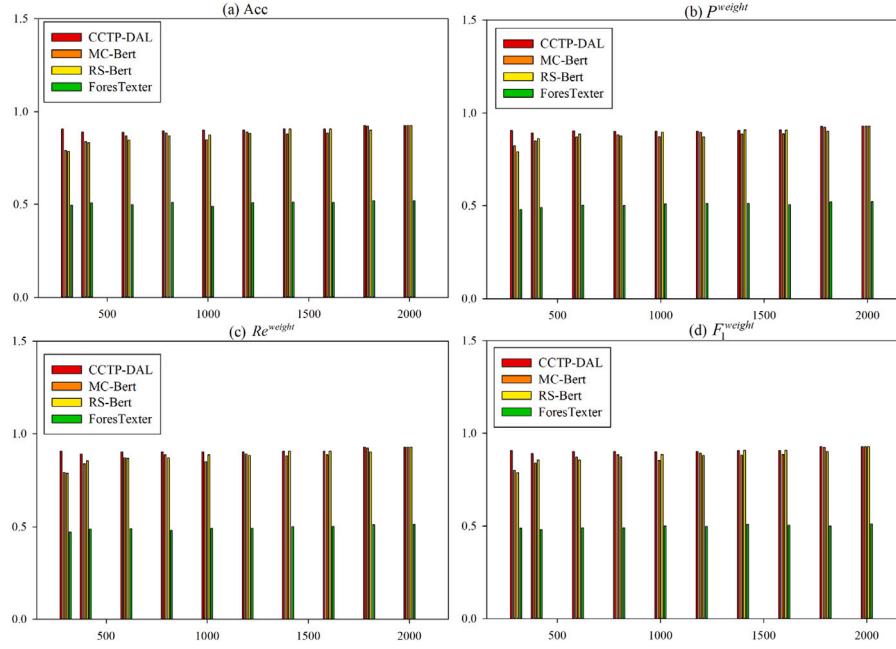


Fig. 5. Performance comparison of various algorithms for PaData.

$X' = \{X'_1, X'_2, \dots, X'_n\}$, and $W_h = \frac{N_h}{N} = \frac{n_h}{n}$ denotes the hierarchical weight of the number of selected text data in the h th layer to the total number of power text data. Then the mean confidence variance of X' is $V(C_{HC}) = \frac{1-f}{n}(S^2 - \sum_{h=1}^L W_h(\bar{C}_h - \bar{C})^2)$.

Proof. According to Theorem 1, we have

$$V(\bar{C}_h) = \frac{1-f_h}{n_h} S_h^2. \quad (14)$$

From the definitions of hierarchical confidence and covariance, we have

$$\begin{aligned} V(C_{HC}) &= V\left(\sum_{h=1}^L W_h \bar{C}_h\right) \\ &= \sum_{h=1}^L W_h^2 V(\bar{C}_h) + 2 \sum_{h=1}^L \sum_{j>h}^L W_h W_j \text{cov}(\bar{C}_h, \bar{C}_j). \end{aligned} \quad (15)$$

And, since the layers in the hierarchical confidence are independent of each other, $\bar{C}_h (h = 1, 2, \dots, L)$ is also independent of each other. We have

$$\text{cov}(\bar{C}_h, \bar{C}_j) = 0. \quad (16)$$

Bringing Eq. (16) into Eq. (15), we have

$$V(C_{HC}) = \sum_{h=1}^L W_h^2 V(\bar{C}_h). \quad (17)$$

Bringing Eq. (14) into Eq. (17), the following equation holds

$$V(C_{HC}) = \sum_{h=1}^L W_h^2 \frac{1-f_h}{n_h} S_h^2. \quad (18)$$

From Theorem 1, we have

$$1-f_h = 1 - \frac{n_h}{N_h} = \frac{N_h - n_h}{N_h}. \quad (19)$$

Then, Eq. (18) can be changed to Eq. (20).

$$\begin{aligned} V(C_{HC}) &= \sum_{h=1}^L W_h^2 \frac{1-f_h}{n_h} S_h^2 \\ &= \sum_{h=1}^L W_h^2 \frac{N_h - n_h}{n_h N_h} S_h^2 \\ &= \sum_{h=1}^L \left(\frac{1}{n_h} - \frac{1}{N_h}\right) W_h^2 S_h^2 \\ &= \sum_{h=1}^L \left(\frac{W_h}{n_h} - \frac{W_h}{N_h}\right) W_h S_h^2 \end{aligned} \quad (20)$$

According to the known conditions, Eq. (20) can be changed to

$$\begin{aligned} V(C_{HC}) &= \sum_{h=1}^L \left(\frac{W_h}{n_h} - \frac{W_h}{N_h}\right) W_h S_h^2 \\ &= \sum_{h=1}^L \left(\frac{1}{n} - \frac{1}{N}\right) W_h S_h^2 = \frac{1-f}{n} \sum_{h=1}^L W_h S_h^2. \end{aligned} \quad (21)$$

From Definition 4, we have

$$\begin{aligned} S_h^2 &= \frac{1}{N_h - 1} \sum_{i=1}^{N_h} (C_i^h - \bar{C}_h)^2 \\ &= \frac{1}{N_h - 1} \sum_{i=1}^{N_h} [(C_i^h - \bar{C}) - (\bar{C}_h - \bar{C})]^2 \\ &= \frac{1}{N_h - 1} \sum_{i=1}^{N_h} (C_i^h - \bar{C})^2 - \frac{1}{N_h - 1} (\bar{C}_h - \bar{C})^2 \end{aligned} \quad (22)$$

We bring Eq. (22) into Eq. (21), the following equation holds

$$\begin{aligned} V(C_{HC}) &= \frac{1-f}{n} \sum_{h=1}^L W_h S_h^2 \\ &= \frac{1-f}{n} \sum_{h=1}^L W_h \left(\frac{1}{N_h - 1} \sum_{i=1}^{N_h} (C_i^h - \bar{C})^2 - \frac{1}{N_h - 1} (\bar{C}_h - \bar{C})^2 \right) \end{aligned} \quad (23)$$

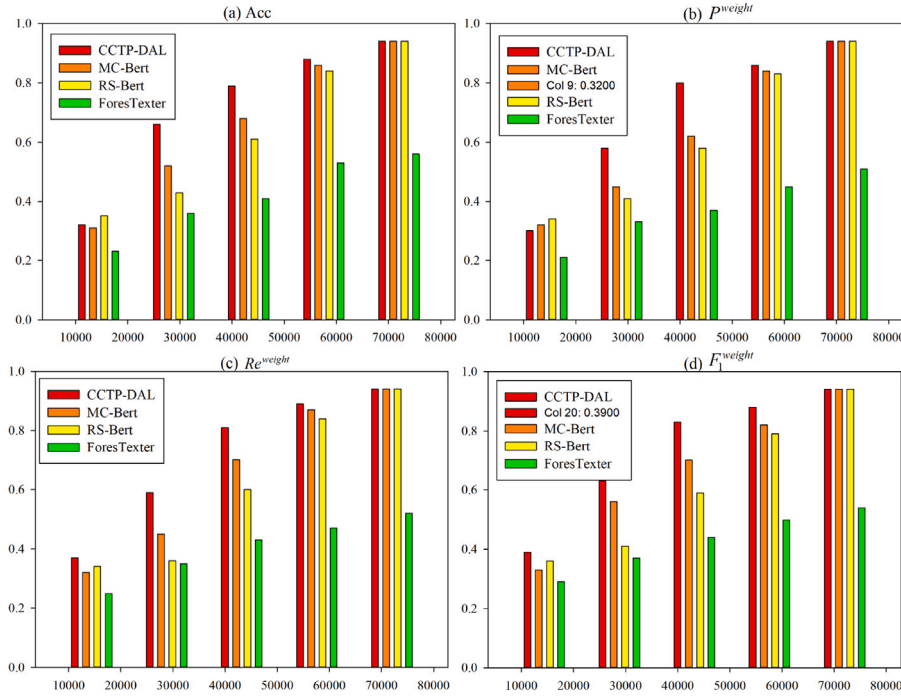


Fig. 6. Performance comparison of various algorithms for THUCNews.

Since $W_h = \frac{N_h}{N}$ and $N_h \gg 1, N \gg 1$, we have

$$V(C_{HC}) = \frac{1-f}{n} \left(\frac{1}{N} \sum_{i=1}^N (C_i - \bar{C})^2 - \sum_{h=1}^L W_h \frac{N_h}{N_h-1} (\bar{C}_h - \bar{C})^2 \right) \quad (24)$$

$$\approx \frac{1-f}{n} (S^2 - \sum_{h=1}^L W_h (\bar{C}_h - \bar{C})^2)$$

From Theorems 1 and 2, we find that when $\bar{C}_h = \bar{C}$, simple random sampling and hierarchical sampling have the same learning ability for sample features. However, in a real text classification scenario, $\bar{C}_h \neq \bar{C}$, and thus $V(\bar{C}) > V(C_{HC})$. The results indicate that hierarchical sampling is better than simple random sampling to obtain more features of text data.

4. Experimental analysis

This section first introduces the experimental data source, the experimental environment, and metrics. Then, we answer the following three questions based on the experiments.

RQ1: Can CCTP-DAL achieve high accuracy in power text classification and reduce annotation cost with a small amount of training data?

RQ2: Can the hierarchical confidence approach improve the performance of complex text classification models and thus expand the application of text classification algorithms in power scenarios?

RQ3: Can CCTP-DAL achieve high text classification accuracy in a short training time, thus reducing the utilization of computational resources by deep learning-based text classification algorithms?

Finally, we verify the reliability and validity of the experimental results by Friedman hypothesis testing.

4.1. Experimental data source

This paper constructs a text classification dataset (called PaData, and can be download from <https://github.com/RGxzz-Lql/Deep-active-learning-classification-data>) for research in the field of electricity based on research papers and reports related to the field of electricity crawled and downloaded from the internet. There are five categories in the

Table 1

Friedman test results.

Dataset	#Instance	#Category	Source
PaData	2401	5	https://github.com/RGxzz-Lql/Deep-active-learning-classification-data
THUCNews	80 000	4	http://thuctc.thunlp.org/

dataset, that include power knowledge graph, power blockchain, power system data distributed storage and recovery, power load forecast and power system information security protection. This dataset contains 2401 pieces of textual data of electricity-related research directions. Among them, there are 505 pieces of knowledge graph data, 481 pieces of power blockchain text data, a total of 328 pieces of power system data distributed storage and recovery data, 645 pieces of power system load forecasting data and 442 items of power system information security protection data. The datasets from different research directions have undergone uniform data cleaning, preprocessing and manual checking to finally form the final text dataset.

To better prove the advantage of our proposed model, we added a benchmark dataset on text classification (called THUCNews, and can be download from <http://thuctc.thunlp.org/>). THUCNews dataset comes from the Natural Language Processing Laboratory of Tsinghua University, and is generated based on the historical data filtered from Sina News RSS feeds between 2005 2011, which contains 740,000 news documents, all in UTF-8 plain text format. The dataset has 14 categories. In the experiment, we selected the four categories of sports, finance, education and entertainment to form the experimental dataset. Each category has 20 000 pieces of data, and the data total 80 000 pieces. In each category, 1000 pieces of data are selected as validation set, 1000 pieces of data are selected as test set, and the remaining 18,000 pieces of data are selected as training data, with a total of 72,000 pieces of training data. Each type of training data is constructed as 20%, 40%, 60%, 80% and 100% of the training data set.

All experimental datasets are shown in Table 1.



Fig. 7. Hierarchy confidence experimental results.

4.2. Experimental setting

In this paper, our experiments are implemented based on the BERT-Base-Chinese model, with 12 transformer blocks, 768 hidden layer states and 12 self-attention heads. Two hundred randomly selected data from the power text data are used as the initial training set to train the base model to calculate the confidence of the unlabeled data. The remaining 1800 data points form the unlabeled data pool. The number of layers in the hierarchy confidence, $H = 5$, $n_i = 360$, are set, and the unlabeled text data are selected for labeling based on the confidence of the layers. The experimental hardware and software environment consisted of i9-10900k, 3.70 GHz CPU, 32 G RAM, NVIDIA GeForce RTX 2080, Python 3.6, PyTorch 3.6 and Cuda 10.1.

4.3. Metrics

Considering the heterogeneous distribution of power text data, this paper selects accuracy and weighted average metrics to evaluate the accuracy of the power text classification algorithm. These metrics are defined using the following functions. TP indicates the number of positive cases predicted correctly, FP indicates the number of negative cases predicted incorrectly, TN indicates the number of negative cases predicted correctly, FN indicates the number of positive cases

predicted incorrectly, and the number of text categories k , where n_i is the number of texts in each category.

Definition 5. Let $ACC = \frac{TP+TN}{TP+TN+FP+FN}$ denote the ratio of the number of all accurately classified text data to the number of all classified text, and call it the accuracy of text classification. The higher the ACC is, the more accurate the text classification is.

Definition 6. Let $P^{weight} = \frac{\sum_{i=1}^k \frac{n_i TP_i}{TP_i + FP_i}}{\sum_{i=1}^k n_i}$ denote the weighted precision at which the positive text is correctly classified in all categories. The higher P^{weight} is, the more precision the text classification is.

Definition 7. Let $Re^{weight} = \frac{\sum_{i=1}^k \frac{n_i TP_i}{2TP_i + TP_i + FN_i}}{\sum_{i=1}^k n_i}$ denote the weighted recall rate at which the positive text is correctly classified in all categories. The higher Re^{weight} is, the more precision the text classification is.

Definition 8. Let $F_1^{weight} = \frac{\sum_{i=1}^k \frac{2n_i TP_i}{2TP_i + TP_i + FN_i}}{\sum_{i=1}^k n_i}$ denote the weighted F_1 for the text classification algorithm. And F_1^{weight} represents the weighted results for F_1 values of different text class. The higher F_1^{weight} is, the better the text classification is.

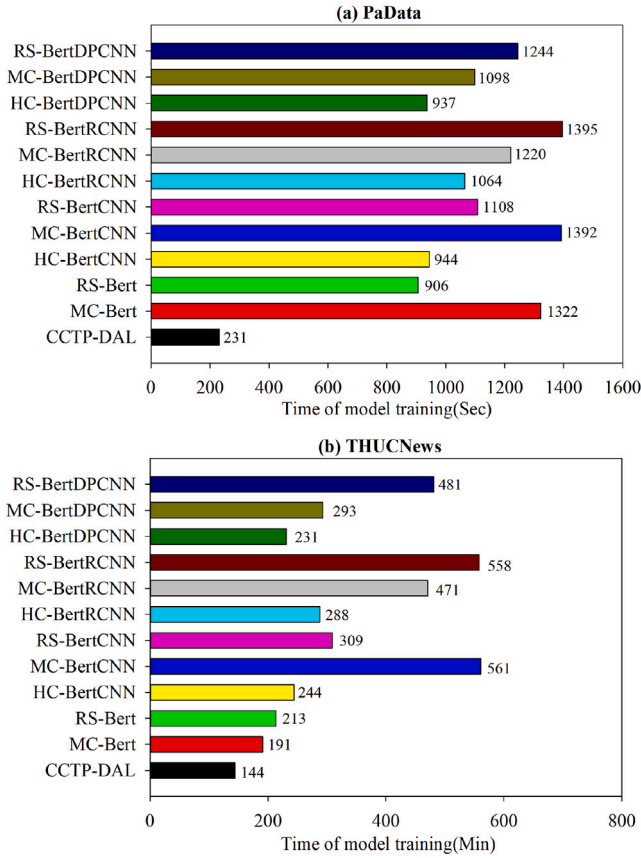


Fig. 8. Comparison of training time between different algorithms.

4.4. Analysis of experimental results

4.4.1. Answering RQ1

In this subsection, we compare the proposed model with the minimum confidence-based Bert text classification (MC-Bert) [35], the random sampling-based Bert text classification (RS-Bert) [36] and Fores-Texter [37] as a way to verify that our model can achieve high classification accuracy with a small number of labels. The experimental results are shown in Figs. 5 and 6.

From these figures, we have the following observations:

- With a small amount of training data (e.g., 300, 400, 600, 800 training data for PaData, and 28 800, 43 200 training data for THUCNews), CCTP-DAL has a significant advantage over MC-Bert, RS-Bert and ForesTexter.
- For the 300 training data in the PaData, compared with MC-Bert, RS-Bert and ForesTexter, the ACC of CCTP-DAL is improved by about 12.64%, 13.77% and 47.26%, respectively. And for the 43 200 training data in the THUCNews, compared with MC-Bert, RS-Bert and ForesTexter, the ACC of CCTP-DAL is improved by about 13.92%, 22.78% and 48.1%, respectively.
- With the increasing size of training data, the ACC of MC-Bert, RS-Bert and ForesTexter is still lower than that of CCTP-DAL, despite the gradual improvement of the ACC .
- For PaData, when the training data size is from 300 to 2000, the improvement of ACC of CCTP-DAL, MC-Bert, RS-Bert, ForesTexter reaches 2.25%, 14.6%, 15.17% and 8.2%, respectively. And, for THUCNews, when the training data size is from 28 800 to 43 200, the improvement of ACC of CCTP-DAL, MC-Bert, RS-Bert, ForesTexter reaches 59.49%, 54.41%, 42.64% and 43.9%. This shows that our model is insensitive to the training data size.

- For PaData, when the size of the training data is from 300 to 800, compared to MC-Bert, RS-Bert and ForesTexter, the P^{weight} of CCTP-DAL is improved by about 9.07%, 12.63%, 47.26%; 4.82%, 3.28%, 45.12%; 3.7%, 1.94%, 44.63%; 1.9%, 2.77%, 44.52%; respectively. And, for THUCNews, when the size of the training data is from 28 800 to 43 200, compared to MC-Bert, RS-Bert and ForesTexter, the P^{weight} of CCTP-DAL is improved by about 22.41%, 29.31% 43.1%; 22.5%, 27.5%, 53.75%; respectively.
- Regardless of the size of the training data, Re^{weight} and F_1^{weight} of CCTP-DAL outperformed MC-Bert, RS-Bert and ForesTexter. Specially, only for PaData, when the size of the training data reaches 1400 or more, Re^{weight} and F_1^{weight} of MC-Bert and RS-Bert are almost equal to those of CCTP-DAL.
- ForesTexter belongs to the typical unbalanced classification algorithms, while the unbalanced features in all experimental datasets are not very obvious, which leads to the disadvantage of Fores-Texter algorithm in each performance index.

Remark. In summary, the experimental results confirm that the CCTP-DAL can guarantee the accuracy of power text classification with a small amount of training data, thus reducing the cost of manual labeling and making it more applicable to the actual scenario of power grid mass text classification.

4.4.2. Verifying RQ2

To verify the improvement of the hierarchical confidence active learning strategy proposed in the paper for the performance of Bert-based models for classification, we combined the hierarchical confidence active learning strategy with BERTCNN [38], BERTRCNN [39], and BERTDPCNN [40] and compared it with the minimum confidence-based active learning strategy [41] and random sampling. Experimental results are shown in Fig. 7.

From Fig. 7, we have the following observations:

- When the training data size is 300, the ACC , P^{weight} , Re^{weight} and F_1^{weight} of text classification based on HC-BertCNN are 84.37%, 85.53%, 84.38% and 85.65%, respectively, which are significantly better than those of text classification based on MC-BertCNN and RS-BertCNN. Until the training data size reaches 2000, the ACC , P^{weight} , Re^{weight} and F_1^{weight} of MC-BertCNN-based text classification and RS-BertCNN-based text classification are comparable to that of HC-BertCNN-based text classification.
- Compared with MC-BertRCNN-based text classification and RS-BertRCNN-based text classification, when the training data size is 300, the ACC , P^{weight} , Re^{weight} and F_1^{weight} of HC-BertCNN-based text classification are improved by about 4.25%, 9.1%; 4.61%, 6.37%; 4.25%, 9.1%; 3.98%, 9.11%, respectively. And, with the increasing size of training data, the gap of ACC , P^{weight} , Re^{weight} and F_1^{weight} between text classification based on HC-BertCNN and text classification based on MC-BertRCNN and RS-BertRCNN becomes smaller and smaller.
- When the training data size reaches 1400, the ACC , P^{weight} , Re^{weight} and F_1^{weight} of text classification based on HC-BertDPCNN reach 92.19%, 92.48%, 92.19% and 92.24%, respectively, which are better than those of text classification based on MC-BertDPCNN and RS-BertDPCNN, and are larger than those of ACC , P^{weight} , Re^{weight} and F_1^{weight} for text classification based on MC-BertDPCNN and RS-BertDPCNN when the training data size is 2000.

Remark. In brief, the hierarchical confidence (HC) can significantly improve the text classification performance with a small amount of labeled data, which greatly saves the number of text data labeled, saves labor cost, and broadens the application of text classification algorithms in practical business scenarios of power system.

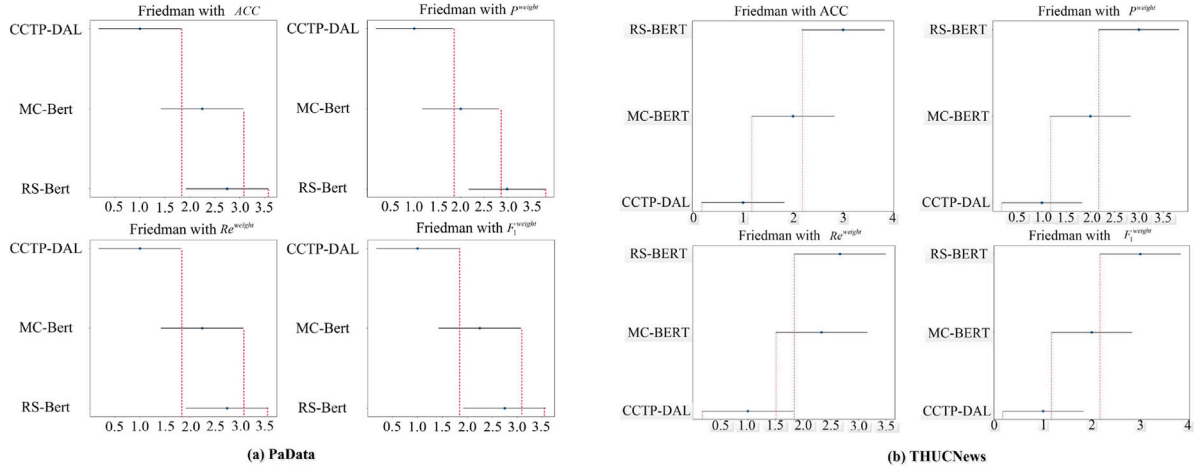


Fig. 9. Friedman side view of the three models based on the Nemenyi test.

4.4.3. Testing RQ3

To validate the advantage of our proposed model in terms of time complexity, we compare the training time required for the different models to achieve 90% classification accuracy. The experimental results are shown in Fig. 8.

From Fig. 8, we can find that compared with other text classification models, for PaData dataset, CCTP-DAL has the smallest training time of only 231 s under the condition that the accuracy of text classification reaches 90% and above, while the text classification based on RS-BertRCNN has the largest training time of 1395 s. And, for THUCNews dataset, CCTP-DAL has the smallest training time of 144 min under the condition that the accuracy of text classification reaches 80% and above, while the text classification based on MC-BertRCNN has the largest training time of 561 min. This is mainly due to the fact that (1) our proposed model and the used hierarchical confidence can obtain a high text classification accuracy with a small amount of training data, and (2) the large size of the THUCNews dataset makes the text categorization model too time-consuming.

4.5. Friedman test

The results of hypothesis testing can be used as a basis for judging the performance of algorithms. The Friedman test is used to determine whether these algorithms perform equally on the same dataset. If the performance is the same, their average sequence values should be comparable. Let K algorithms be compared on N datasets, and r_i denote the mean ordinal value of the i th algorithm. Then the mean and variance of r_i are $(K+1)/2$ and $(K^2+1)/12N$, respectively, and the variable $\Gamma_F = \frac{(N-1)\Gamma_{\chi^2}}{N(K-1)-\Gamma_{\chi^2}}$ obeys the F distribution with degrees of freedom $k-1$ and $(k-1)(N-1)$, where $\Gamma_{\chi^2} = \frac{12N}{K(K+1)}(\sum_{i=1}^K r_i^2 - \frac{K(K+1)^2}{4})$.

To test the credibility of the experimental results that the hierarchical confidence proposed in this paper has higher classification accuracy than MC-Bert and RS-Bert models on a small amount of training data, for PaData and THUCNews datasets, we perform Friedman test and Nemenyi follow-up test on the experimental results of the above three models on training data sizes of 300, 400, 600, 800 and 28 800, 43 200, respectively. The result of Friedman test is shown in Table 2.

From Table 2, we know that the p -values of all metrics are less than 0.05, which indicates that the hypothesis of “all algorithms have the same performance” is rejected. The results indicate that the algorithms have significantly different performance. At this point, we further distinguish the performance of various algorithms by the Nemenyi follow-up test. The experimental results are shown in Fig. 9.

From the above figure, we have the following conclusions:

Table 2

Friedman test results.

PaData		THUCNews	
Index	p-value	Index	p-value
Acc	0.0327	Acc	0.042
p_{weight}	0.0425	p_{weight}	0.043
Re_{weight}	0.0387	Re_{weight}	0.039
$F1_{weight}$	0.0387	$F1_{weight}$	0.042

- Each of the three algorithms differed from each other under four comparative metrics.
- For Acc , p_{weight} , Re_{weight} and $F1_{weight}$, there is no overlapping region between CCTP-DAL and RS-Bert models, indicating a significant difference in performance between these two models.
- Although the CCTP-DAL and MC-Bert models have overlap, our proposed CCTP-DAL significantly outperforms MC-Bert in terms of the above experimental results of all metrics.

5. Conclusions

In this paper, we investigate the problem of how to achieve an accurate classification of a large number of unlabeled short texts in the specialist electricity domain using a small amount of labeled text data. We propose a deep active learning-based power text classification method. This paper concludes with the following results:

1. An active learning method with hierarchical confidence is proposed to divide the sample data into several levels. The implementation divides the overall sample data, with large internal variability, into multiple smaller overall data, with smaller internal variability, which improves the new features of the training data and can effectively improve the generalization ability of the model.
2. The power texts based on the Bert model are efficiently encoded. The combination of the mainstream BERT natural language processing model and an active learning mechanism effectively reduces the dependence of the deep learning methods on massive amounts of data.
3. Based on the experimental results, it is demonstrated that the basic deep active learning power text classification method proposed in this paper can use as little labeled text data as possible with a guaranteed accuracy.

CRediT authorship contribution statement

Song Deng: Conceptualization, Methodology, Software, Writing – original draft, Formal analysis, Funding acquisition, Data curation, Writing – review & editing. **Qianliang Li:** Conceptualization, Methodology, Software, Writing – original draft, Formal analysis, Funding acquisition, Data curation, Writing – review & editing. **Renjie Dai:** Resources, Investigation. **Siming Wei:** Resources, Investigation. **Di Wu:** Data curation, Writing – review & editing, Investigation. **Yi He:** Data curation, Writing – review & editing, Investigation. **Xindong Wu:** Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 51977113, 62293500, 62293501 and 62293505), State Grid Shanghai Municipal Electric Power Company Technology Projects (No. B30935210005).

References

- [1] A.-H. Mohsenian-Rad, V.W. Wong, J. Jatskevich, R. Schober, A. Leon-Garcia, Autonomous demand-side management based on game-theoretic energy consumption scheduling for the future smart grid, *IEEE Trans. Smart Grid* 1 (3) (2010) 320–331.
- [2] G. Xylomenos, C.N. Ververidis, V.A. Siris, N. Fotiou, C. Tsilopoulos, X. Vasilakos, K.V. Katsaros, G.C. Polyzos, A survey of information-centric networking research, *IEEE Commun. Surv. Tutorials* 16 (2) (2013) 1024–1049.
- [3] Z. Wang, J. Zhang, J. Feng, Z. Chen, Knowledge graph embedding by translating on hyperplanes, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 28, 2014.
- [4] Y. Lin, Z. Liu, M. Sun, Y. Liu, X. Zhu, Learning entity and relation embeddings for knowledge graph completion, in: *Twenty-Ninth AAAI Conference on Artificial Intelligence*, 2015.
- [5] J.Y. Lee, F. Démoncourt, Sequential short-text classification with recurrent and convolutional neural networks, 2016, arXiv preprint arXiv:1603.03827.
- [6] X. Zhang, J. Zhao, Y. LeCun, Character-level convolutional networks for text classification, *Adv. Neural Inf. Process. Syst.* 28 (2015).
- [7] J. Liu, H. Ma, X. Xie, J. Cheng, Short text classification for faults information of secondary equipment based on convolutional neural networks, *Energies* 15 (7) (2022) 2400.
- [8] F. Zhao, C. Jia, W. Li, J. Hao, J. Yin, L. Pan, Y. Yang, A text classification method of power grid assets based on improved FastText, in: *Third International Conference on Computer Science and Communication Technology (ICCSCT 2022)*, Vol. 12506, SPIE, 2022, pp. 49–56.
- [9] W. Ding, S. Jin, Y. Ren, F. Liu, Research on classification method of business requirement text based on deep learning, in: *Proceedings of the 7th International Conference on Cyber Security and Information Engineering*, 2022, pp. 222–226.
- [10] X. Jing, Z. Wu, L. Zhang, Z. Li, D. Mu, Electrical fault diagnosis from text data: A supervised sentence embedding combined with imbalanced classification, *IEEE Trans. Ind. Electron.* (2023) 1–10.
- [11] Q. Meng, Y. Song, J. Mu, Y. Lv, J. Yang, L. Xu, J. Zhao, J. Ma, W. Yao, R. Wang, et al., Electric power audit text classification with multi-grained pre-trained language model, *IEEE Access* 11 (2023) 13510–13518.
- [12] F. Sebastiani, Machine learning in automated text categorization, *ACM Comput. Surv. (CSUR)* 34 (1) (2002) 1–47.
- [13] S. Minaee, N. Kalchbrenner, E. Cambria, N. Nikzad, M. Chenaghlu, J. Gao, Deep learning-based text classification: a comprehensive review, *ACM Comput. Surv.* 54 (3) (2021) 1–40.
- [14] H. Zhang, D. Li, Naïve Bayes text classifier, in: *2007 IEEE International Conference on Granular Computing (GRC 2007)*, IEEE, 2007, p. 708.
- [15] M.A. Klopotek, M. Woch, Very large Bayesian networks in text classification, in: *International Conference on Computational Science*, Springer, 2003, pp. 397–406.
- [16] H. Kim, P. Howland, H. Park, N. Christianini, Dimension reduction in text classification with support vector machines, *J. Mach. Learn. Res.* 6 (1) (2005).
- [17] J. Su, H. Zhang, A fast decision tree learning algorithm, in: *AAAI*, Vol. 6, 2006, pp. 500–505.
- [18] Z. Yong, L. Youwen, X. Shixiong, An improved KNN text classification algorithm based on clustering, *J. Comput.* 4 (3) (2009) 230–237.
- [19] L. Shi, X. Ma, L. Xi, Q. Duan, J. Zhao, Rough set and ensemble learning based semi-supervised algorithm for text classification, *Expert Syst. Appl.* 38 (5) (2011) 6300–6306.
- [20] Y. Zhang, R. Jin, Z.-H. Zhou, Understanding bag-of-words model: a statistical framework, *Int. J. Mach. Learn. Cybern.* 1 (1) (2010) 43–52.
- [21] Q. Zhang, R. Zheng, Z. Zhao, B. Chai, J. Li, A textcnn based approach for multi-label text classification of power fault data, in: *2020 IEEE 5th International Conference on Cloud Computing and Big Data Analytics (ICCCBDA)*, IEEE, 2020, pp. 179–183.
- [22] T. He, W. Huang, Y. Qiao, J. Yao, Text-attentional convolutional neural network for scene text detection, *IEEE Trans. Image Process.* 25 (6) (2016) 2529–2541.
- [23] H. Peng, J. Li, S. Wang, L. Wang, Q. Gong, R. Yang, B. Li, S.Y. Philip, L. He, Hierarchical taxonomy-aware and attentional graph capsule RCNNs for large-scale multi-label text classification, *IEEE Trans. Knowl. Data Eng.* 33 (6) (2019) 2505–2519.
- [24] Q. Yang, J. Jiang, X. Feng, M. Li, J. He, L. Zhang, Text classification method for analysing accidents in power system operation, in: *2021 6th Asia Conference on Power and Electrical Engineering (ACPEE)*, IEEE, 2021, pp. 1526–1530.
- [25] K. Chen, D. Nan, Y. Sun, K. Wang, Text mining of power secondary equipment based on BiLSTM-attention, in: *2020 Chinese Control and Decision Conference (CCDC)*, IEEE, 2020, pp. 709–714.
- [26] J.D.M.C.K. Lee, K. Toutanova, Pre-training of deep bidirectional transformers for language understanding, 2018, arXiv preprint arXiv:1810.04805.
- [27] Z. Lin, D. Lei, Y. Han, G. Wang, W. Deng, Y. Huang, Siamese BERT model with adversarial training for relation classification, in: *2020 IEEE International Conference on Knowledge Graph (ICKG)*, IEEE, 2020, pp. 291–296.
- [28] D. Croce, G. Castellucci, R. Basili, GAN-BERT: Generative adversarial learning for robust text classification with a bunch of labeled examples, in: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 2114–2119.
- [29] D. Chai, W. Wu, Q. Han, F. Wu, J. Li, Description based text classification with reinforcement learning, in: *International Conference on Machine Learning, PMLR*, 2020, pp. 1371–1382.
- [30] C. WEN, C. ZENG, J. REN, Y. ZHANG, Patent text classification based on ALBERT and bidirectional gated recurrent unit, *J. Comput. Appl.* 41 (2) (2021) 407.
- [31] X. Dong, S. Deng, D. Wang, A short-term power load forecasting method based on k-means and SVM, *J. Ambient Intell. Humaniz. Comput.* (2021) 1–15.
- [32] M.L. Di Silvestre, P. Gallo, J.M. Guerrero, R. Musca, E.R. Sanseverino, G. Sciumè, J.C. Vázquez, G. Zizzo, Blockchain for power systems: Current trends and future applications, *Renew. Sustain. Energy Rev.* 119 (2020) 109585.
- [33] P. Gao, M. Wang, S.G. Ghiocel, J.H. Chow, B. Fardanesh, G. Stefopoulos, Missing data recovery by exploiting low-dimensionality in power system synchrophasor measurements, *IEEE Trans. Power Syst.* 31 (2) (2015) 1006–1013.
- [34] D. Hendrycks, K. Gimpel, Gaussian error linear units (gelus), 2016, arXiv preprint arXiv:1606.08415.
- [35] L. Zhang, L. Zhang, An ensemble deep active learning method for intent classification, in: *Proceedings of the 2019 3rd International Conference on Computer Science and Artificial Intelligence*, 2019, pp. 107–111.
- [36] C. Sun, X. Qiu, Y. Xu, X. Huang, How to fine-tune bert for text classification? in: *China National Conference on Chinese Computational Linguistics*, Springer, 2019, pp. 194–206.
- [37] Q. Wu, Y. Ye, H. Zhang, M.K. Ng, S.-S. Ho, ForestText: An efficient random forest algorithm for imbalanced text categorization, *Knowl.-Based Syst.* 67 (2014) 105–116.
- [38] J. Dong, F. He, Y. Guo, H. Zhang, A commodity review sentiment analysis based on BERT-CNN model, in: *2020 5th International Conference on Computer and Communication Systems (ICCCS)*, IEEE, 2020, pp. 143–147.
- [39] P. Ni, Y. Li, G. Li, V. Chang, Natural language understanding approaches based on joint task of intent detection and slot filling for IoT voice interaction, *Neural Comput. Appl.* 32 (20) (2020) 16149–16166.
- [40] Y.-J. Li, H.-J. Zhang, W.-M. Pan, R.-J. Feng, Z.-Y. Zhou, Microblog rumor detection based on bert-DPCNN, in: *Artificial Intelligence in China*, Springer, 2021, pp. 524–530.
- [41] M. Goudjil, M. Koudil, M. Bedda, N. Hoggali, A novel active learning method using SVM for text classification, *Int. J. Autom. Comput.* 15 (3) (2018) 290–298.